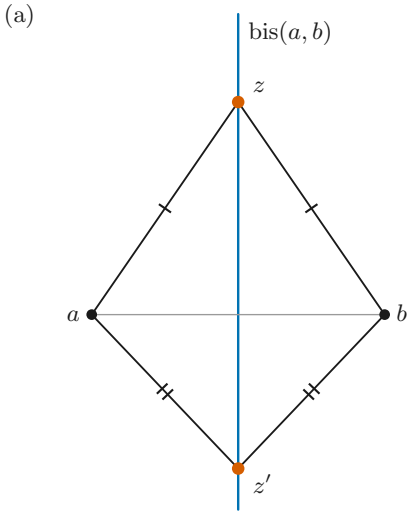
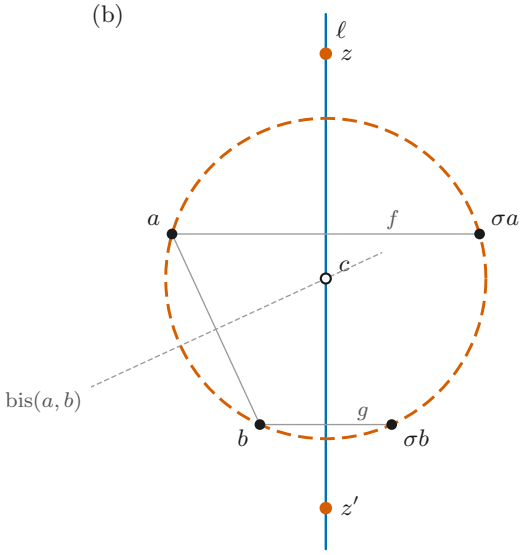
a third apex would be a third point of P on the line $\mathrm{bis}(a, b)$